

Assignment: Library Management System API (NestJS + PostgreSQL)

Objective

Build a backend API for a **Library Management System** using:

- NestJS
- PostgreSQL
- TypeORM

The goal is to implement:

- ✓ CRUD operations
 - ✓ One-to-One, One-to-Many, Many-to-Many relationships
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Modules to Create

- Member
 - Book
 - Author
 - MembershipCard
 - Borrowing
-

Database Relationships

✓ One-to-One

- One **Member** has one **MembershipCard**
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✓ One-to-Many

- One **Author** can write many **Books**
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✓ Many-to-Many

- Members can borrow many Books
 - Books can be borrowed by many Members
 - 👉 (Use a **Borrowing** table)
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Requirements

Member

- Create member
 - Get all members
 - Get single member
-

Membership Card (1-1)

- Issue card to a member
 - Get card by member
-

Author

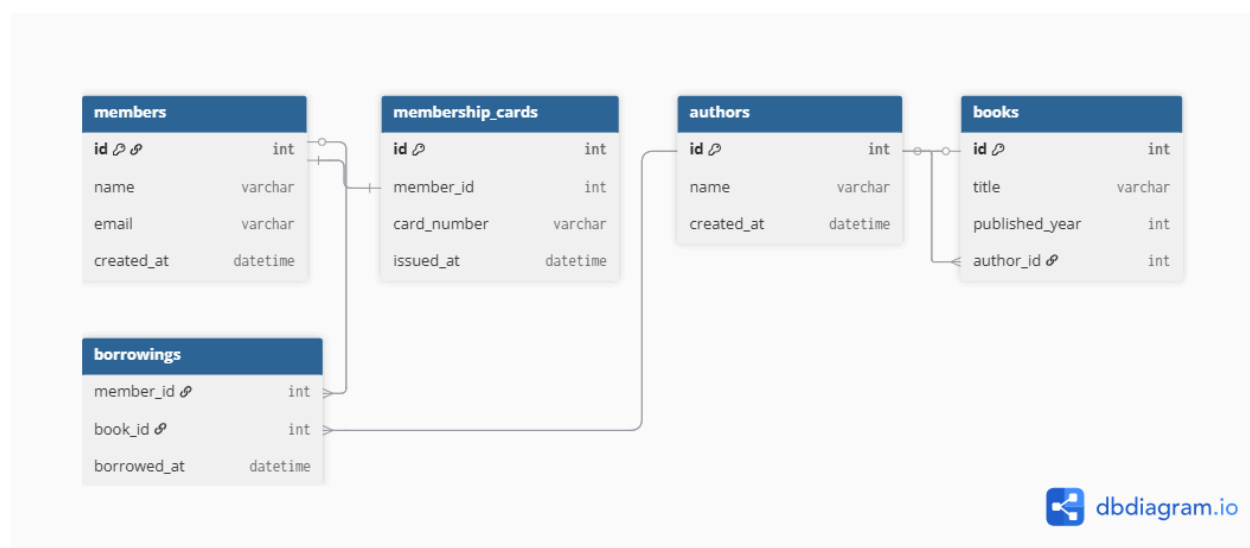
- Create author
 - Get all authors
-

Book

- Create book with author
 - Get all books
 - Get book with author
-

Borrowing (Many-to-Many)

- Borrow a book



Good luck!